

Full Stack Development with MERN

Project Documentation

1. Introduction

- **Project Title :** ShopEZ E-Commerce Application
- **Team Members:** Kancharla Ushasree (Individual Project)

2. Project Overview

• **Purpose :** The ShopeZ E-Commerce Application is designed to provide a seamless online shopping experience for users. It enables customers to browse products, add items to their cart, make secure payments, and track orders. Administrators can manage products, inventory, and orders through a dedicated dashboard, ensuring smooth operations and efficient order fulfillment.

Features: User registration and login
Product browsing and search
Add to cart and checkout
Secure payment processing
Order tracking and history
Admin dashboard for product and order management

3. Architecture

- **Frontend:** React.js - builds a responsive, interactive interface for product browsing, cart management, and checkout.
- **Backend:** Node.js and Express.js — handles RESTful APIs, authentication, and business logic.
- **Database:** MongoDB Atlas — stores user, product, cart, and order data using Mongoose schemas for efficient data management.

4. Setup Instructions

- **Prerequisites:**
Node.js, MongoDB, npm, and Visual Studio Code are required to run the project.

Installation : Clone the repository.
Run `npm install` to install dependencies.
Configure `.env` file with database URI and JWT secret.
Start servers:
Frontend → `npm run dev`
Backend → `npm start`

3. Folder Structure

- **Client:** React.js frontend containing components, pages, services, and routing for the

user interface.

- **Server:** Node.js and Express.js backend containing routes, controllers, models, middleware, and configuration files.

6. Running the Application

- o **Frontend:** npm run dev in the frontend directory.
- o **Backend:** npm start in the server directory.

7. API Documentation

The backend exposes RESTful APIs for:

User Authentication — registration, login, JWT token generation.

Product Management — CRUD operations for products.

Cart & Checkout — add, update, and remove items from cart; process orders.

Order Tracking — retrieve order status and history.

Admin Control — manage inventory and orders.

8. Authentication : Implemented using **JWT (JSON Web Tokens)**. After login, users receive a token stored in an HTTP-only cookie, which secures access to protected routes and ensures safe session management..

9. User Interface :

The ShopeZ UI includes:

Home Page

Product Catalogue Page

Login Page

Registration Page

Shopping Cart Page

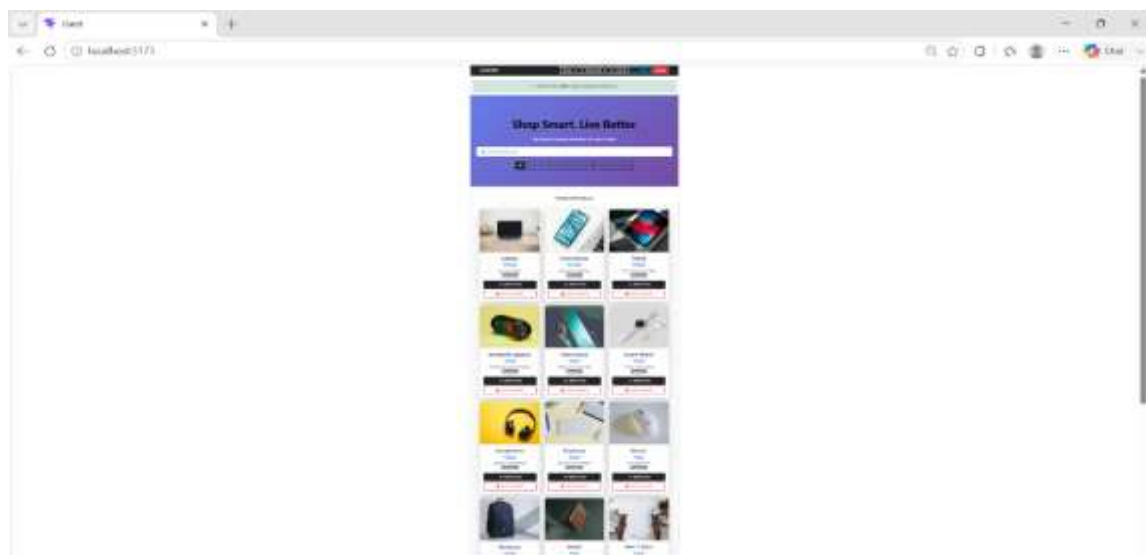
Checkout Page

Order Tracking Page

Admin Dashboard

Each page is designed for clarity, responsiveness, and ease of navigation.

9. Home Page :



Login Page

SHOPEZ

Home

Login

Register

Login

ushasree23k@gmail.com

.....

Login

Register Page

SHOPEZ

Home

Login

Register

Register

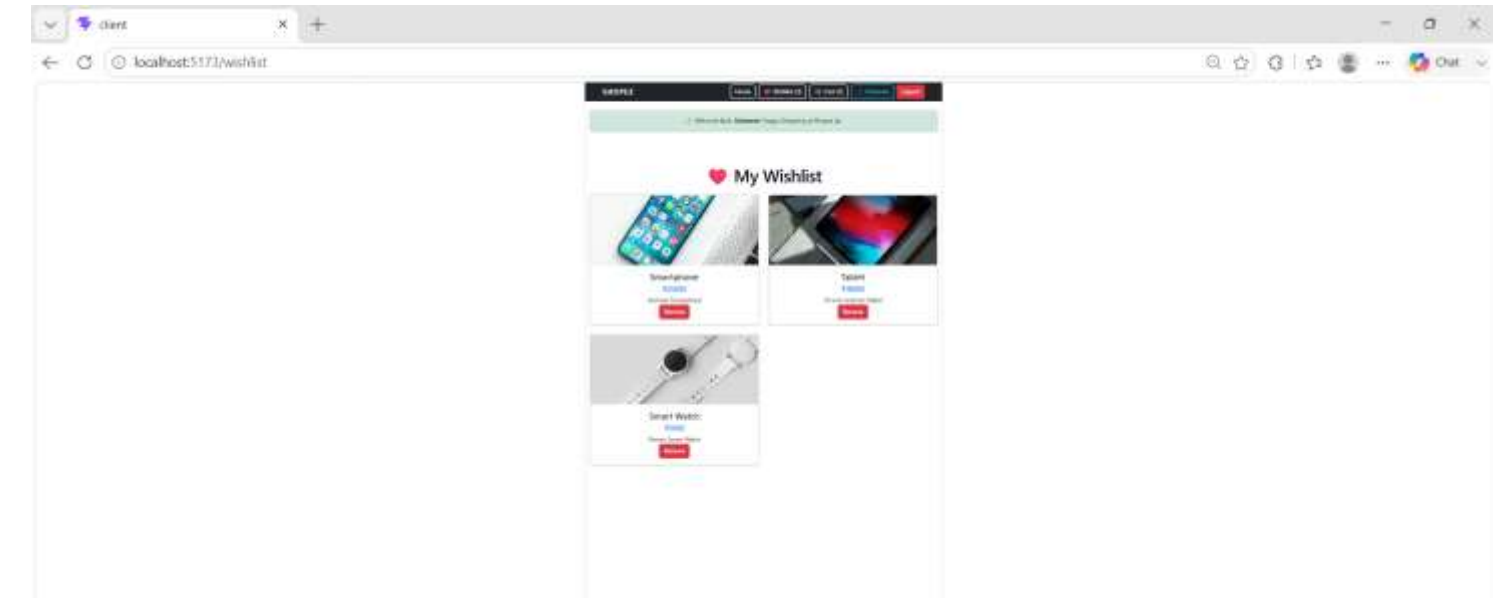
Usha

ushasree23k@gmail.com

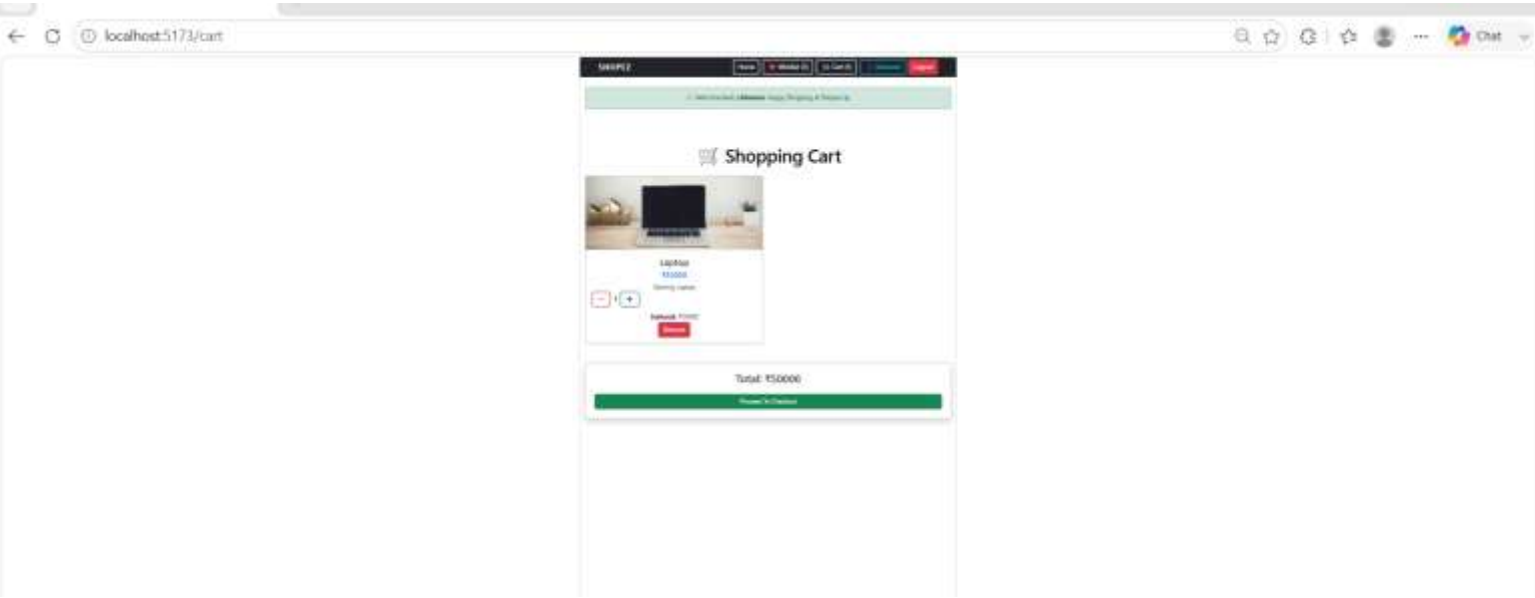
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Register

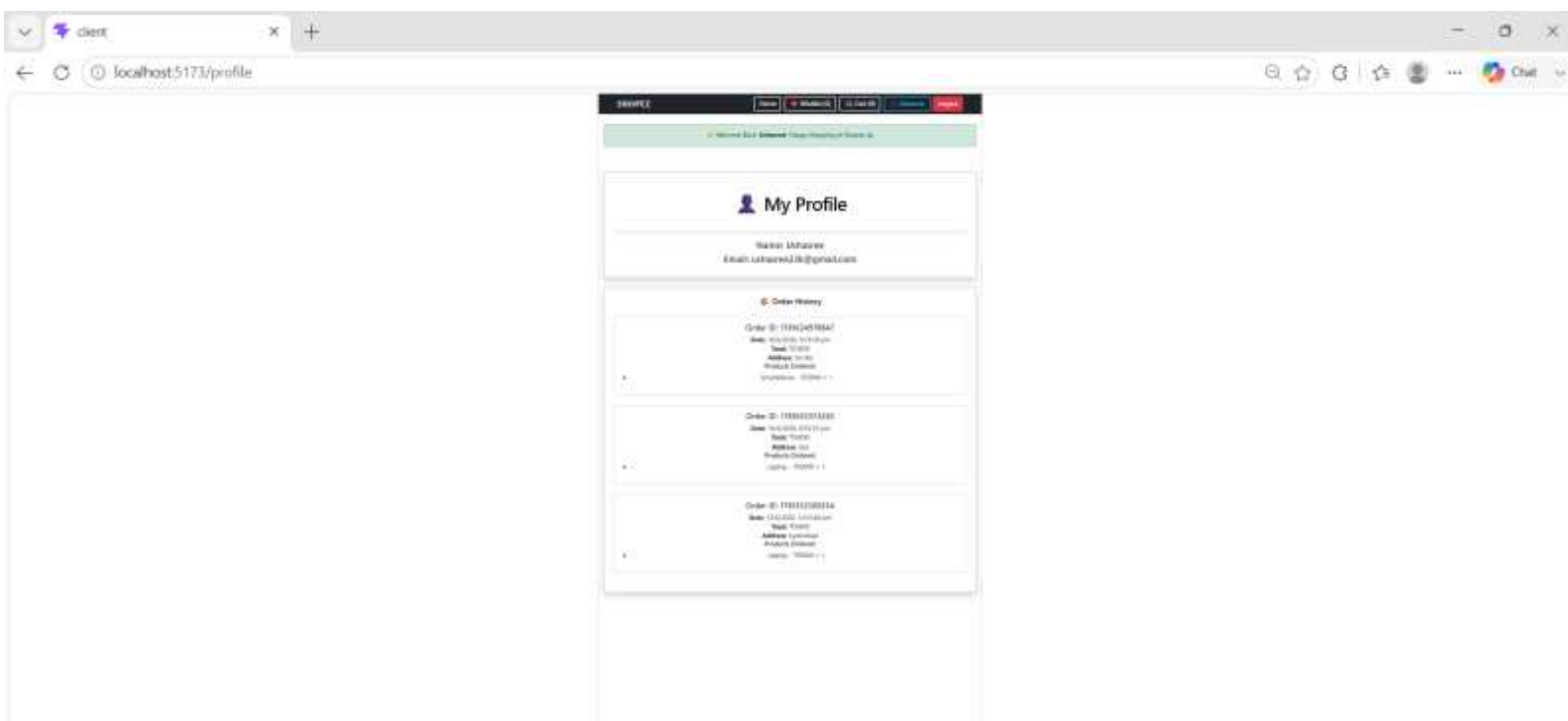
Wishlist Page



Cart Page



User Dashboard



10. Testing : Testing was performed using **Thunder Client / Postman** to verify API endpoints and responses. Manual testing validated registration, login, product listing, cart operations, checkout, and order tracking. Authentication and authorization were tested to ensure secure access and smooth user experience.

11. Screenshots or Demo : The application was successfully developed and tested locally using React.js, Node.js, Express.js, and MongoDB. Screenshots of major modules — Registration, Login, Product Catalogue, Wishlist Page, Cart, Checkout, Order Tracking — demonstrate the complete workflow.

12. Known Issues

Minor UI enhancements and performance optimizations can be added.

Future versions may include advanced features like coupon codes, wishlist management, and multi-vendor support.

No major bugs identified; all core functionalities work as expected.

13. Future Enhancements

Integration of **email notifications** for order updates.

Analytics dashboard for sales tracking.

Real-time chat support for customers.

Mobile application version for broader accessibility.

AI-based recommendation system for personalized shopping.